

This provides a lens on grokking: as routing evolves from random to structured, the effective dimension collapses, triggering improvements in generalization. Intuitively, this collapse corresponds to a simplification of routing pathways: the kernel spectrum becomes more concentrated, reducing the hypothesis space and mitigating overfitting. The full theoretical setup, assumptions, and proof are detailed in Appendix C, while Appendix C.4 empirically confirms that effective dimension is tightly aligned with routing edit distance, establishing pathway complexity as a key driver of MoE generalization.

5 Conclusion

This paper provides the first empirical evidence that, in large-scale LLM pretraining, generalization can emerge well after memorization has been achieved. Through an analysis of routing dynamics in a 7B-parameter MoE model, we propose two novel pathway-based metrics that accurately track generalization without requiring finetuning or external validation sets. Our findings reveal a transition from memorization to generalization, marked by increased pathway similarity and consistency, and supported by theoretical connections between routing structure and generalization bounds. These insights offer a foundation for more transparent and reliable monitoring of generalization in large-scale models. Future work will extend these pathway-based metrics beyond MoE, by constructing analogous “virtual pathways” in dense models, moving toward a unified framework for tracking generalization in foundation models.

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